Policy

Ward based management of non-traumatic subarachnoid haemorrhage

Key messages

- This policy provides guidance for the management of patients with a non-traumatic subarachnoid haemorrhage in the acute setting.
- This guidance relates to patients undergoing a ward based level of care as distinct from an ICU/HDU setting.
- Guidance is provided for management in the pre-treatment and post-treatment phase.
- Guidance for the recognition of complications and subsequent acute management is also provided.

1 Scope

Trust-wide.

2 Purpose

This document provides guidance of patient management for ward-based patients who have experienced a non-traumatic subarachnoid haemorrhage. Details are provided for pre and post coiling/ clipping procedures and how to recognise and manage complications of the procedure.

3 Definitions

ABC	airway, breathing and circulation
BD	twice daily
BP	blood pressure
CT	computer tomography
FBC	full blood count
G&S	group and screen
GCS	Glasgow coma score
IV	intravenous
LMWH	low molecular weight heparin
MAP	mean arterial pressure
NCCU	neurosciences critical care unit
NSAID	non-steroidal anti-inflammatory drug
PO	oral

QDS	four times daily
SAH	subarachnoid haemorrhage
SBP	systolic blood pressure
SIADH	syndrome of inappropriate antidiuretic hormone
SpR	specialist registrar
TDS	tThree times daily
U&E	urea and electrolytes
VTE	venous thromboembolism
X-match	cross matching bloods

4 Pre-coiling / pre-clipping management

- Hourly neurological observations[26]
- Bed rest with no limited elevation [25-26], can sit out to use the commode only
- 3 litres of IV sodium chloride 0.9% per 24 hours [1, 2, 26], caution in elderly or in patients with a cardiac history.
- Aim for euvolaemia.
- Nimodipine 60 mg 4 hourly for 21 days from ictus [3].
- Bloods on admission: FBC, U&Es, coagulation, G&S, X-match (if for theatre)
    - U&Es daily
- Regular analgesia:
    - IV paracetamol 1g TDS and oral codeine 30-60mg QDS [4-26]
    - Avoid NSAIDs [4]
- Blood pressure management:
    - BP management following SAH before the aneurysm is secured involves balancing the competing risks of re-bleeding (due to hypertension) and infarction (due to hypotension). Following SAH, the range of BP values in which cerebral autoregulation can occur becomes compromised which means that brain perfusion becomes more dependent on blood pressure.
    - SBP 90–160 mmHg [5-7] or MAP 105-120 mmHg [8]
- Aim to avoid sharp drops in BP [9]
- If SBP <90 mmHg, consider nimodipine 30 mg 2 hourly.
- Glucose control: 6-10 mmol/L [10–13]
- Laxatives / stool softeners:
    - Lactulose 10 mg BD
- VTE prophylaxis: intermittent pneumatic compression devices [7, 14–16]
- Oxygen saturations maintained >96% with supplemental oxygen as required.
- Patient should be escorted by competent member of staff for transfer to another department eg Angio or CT scan.

5 Post-coiling/ post-clipping management

- Prophylactic LMWH (dalteparin) to start 24 hours post-procedure.
- Avoid aggressive BP control unless SBP >200 [7], MAP exceeds 130 mmHg[16].
    - U&Es daily.
    - Gradual return to no restrictions on mobility.

  - Patient can sit out day 1 post-coiling/clipping.
- Monitor for complications.

6 Recognition and Management of Complications

6.1 Complication: Vasospasm (46% [17])

- Decreasing of fluctuating level of consciousness.
- Focal neurological signs.
- Increasing headache.

6.1.1 Management: Vasospasm

- ABCs, IV fluid bolus.
- Aim to increase MAP.
- CT perfusion scan [18]
- Discuss with SpR and NCCU.

6.2 Complication: Re-bleed (7% [17])

- Sudden reduction in GCS.
- Headache increasing in severity.
- Periods of raised BP associated.

6.2.1 Management: Re-bleed

- ABCs.
- CT head.
- Discuss with SpR and NCCU.

6.3 Complication: Hydrocephalus (20% [19])

- Gradual decline in GCS.
- Increasing headache.
- Nausea and vomiting.

6.3.1 Management: Hydrocephalus

- ABCs.
- CT head.
- Surgical intervention.

6.4 Complication: Seizure (3-22% [20-21])

6.4.1 Management: Seizure

- ABCs.

- U&E: sodium level.
- CT head (to exclude an alternative cause).
- IV levetiracetam 1 g loading dose.
- PO levetiracetam 500 mg BD maintenance dose.

6.5 Complication: Hyponatraemia (30-56% [22-24])

- Cerebral salt wasting: Hypovolaemic with polyuria.
  - IV sodium chloride 0.9%.
  - Aim for euvolaemia.
- SIADH: Euvolaemic with oliguria
  - Fluid restriction.
  - Titrate to sodium level.

7 Monitoring compliance with and the effectiveness of this document

The key standards of this policy document that will be monitored are:

- 100% of patients undergoing VTE assessment and where appropriate, treated with intermittent pneumatic compression devices
- 100% of patients prescribed Nimodipine at the correct dose within four hours of admission
- 100% of patients with blood glucose recorded within 12 hours of admission and efforts made to control as appropriate (with reference to Section 4 above)
- 100% of patients with blood test taken within 24 hours of admission, in particular the electrolyte levels (noting the importance of sodium monitoring)

These KPIs will be initially assessed four months following the introduction of this policy. This assessment will be undertaken by staff from the junior neurosciences rota (staff grades varying from FY2-ST1), under consultant supervision. Once the initial assessment has taken place, the timing of subsequent monitoring will be decided. The results of this monitoring will be reviewed by the directorate audit lead and any changes implemented as appropriate. This may include further teaching on the topic, or the development of Epic smartphrases/ reminders. These changes will be implemented under the guidance of the directorate audit lead and the effects of any changes implemented will be assessed by re-audit.

8 References

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